

TEMPLE UNIVERSITY
FOX SCHOOL OF BUSINESS

AS 3502 / AS 5103 Advanced Long-Term Actuarial Modeling
Chapter 8 Problem Set

DR. KRUPA S. VISWANATHAN

1. You are given a three-state model.

State 0: Healthy
State 1: Disabled
State 2: Dead

You are given the following transition intensities for all t :

- (i) $\mu_x^{01}(t) = 0.02$; $\mu_x^{02}(t) = 0.015$; $\mu_x^{12}(t) = 0.025$
(ii) All other transition intensities are equal to 0.

Jack (x), who is currently Healthy, purchases a whole life insurance that pays a benefit of 125,000 at the moment of death, regardless of which state he was in immediately prior. You are given a force of interest of 1.5%.

Calculate the net single premium for this policy.

2. You are given a three-state model: Healthy (state 0); Sick (state 1), and Dead (state 2), and the following transition intensities for (x) for all t :

- (i) $\mu_x^{01}(t) = 0.022$; $\mu_x^{02}(t) = 0.024$
(ii) $\mu_x^{12}(t) = 0.031$; $\mu_x^{10}(t) = 0.013$
(iii) All other transition intensities are equal to 0.

Jane (x) is currently Healthy.

Using Kolmogorov's forward equations with step $h = 1/2$, calculate the probability that Jane is Healthy at the end of one year.

3. You are given a three-state model and the following transition intensities for (x) for all t :

- (i) $\mu_x^{01}(t) = 0.012$; $\mu_x^{02}(t) = 0.014$
(ii) $\mu_x^{12}(t) = 0.015$; $\mu_x^{10}(t) = 0.008$
(iii) All other transition intensities are equal to 0.

(x) is currently in state 0.

Using Kolmogorov's forward equations with step $h = 1/3$, calculate the probability that (x) will be in state 1 at the end of 8 months.

4. You are given a three-state model and the following transition intensities for (x) for all t :

(i) $\mu_x^{01}(t) = 0.015$; $\mu_x^{02}(t) = 0.018$

(ii) $\mu_x^{12}(t) = 0.024$; $\mu_x^{10}(t) = 0.016$

(iii) All other transition intensities are equal to 0.

(x) is currently in state 0.

- (a) Write the Kolmogorov forward differential equations with the associated boundary conditions for this model for:

(i) ${}_t p_x^{00}$

(ii) ${}_t p_x^{01}$

- (b) Using Euler's method and Kolmogorov's forward equations with step $h = 1/4$, calculate the probability that (x) will be in state 0 at the end of 6 months.

5. You are given a three-state model.

State 0: Active

State 1: Disabled

State 2: Dead

You are given the following transition intensities for all t :

(i) $\mu_{45}^{01}(t) = 0.019$; $\mu_{45}^{02}(t) = 0.012$; $\mu_{45}^{12}(t) = 0.035$

(ii) All other transition intensities are equal to 0.

Jenny (45) is currently Active and purchases a whole life insurance that pays a benefit of 60,000 at the moment of death. You are given a force of interest of 1%.

Calculate the net single premium for this policy.

6. You are given a three-state model about the OWL Company, which offers a retirement plan and death benefits to its employees.

State 0: Active employee

State 1: Retired

State 2: Dead

You are given the following transition intensities for all t :

- (i) $\mu_x^{01}(t) = 0.05$; $\mu_x^{02}(t) = 0.035$; $\mu_x^{12}(t) = 0.0425$
(ii) $\mu_x^{10}(t) = 0$

As the actuary for the OWL Company, you are interested in transitions over the next 5 years for (x). Create this 5-year transition matrix by calculating each transition probability independently.

7. For a multi-state model with 3 possible states, an employee of the TU Company can be in one of the following states at the beginning of each year: Active (state 0), Retired (state 1), Dead (state 2).

A currently Active TU employee, Jennifer, just turned 50 years old. You are given the following table of her annual transition probabilities for $k = 0, 1, 2, 3$.

State i	p_{50+k}^{i0}	p_{50+k}^{i1}	p_{50+k}^{i2}
Active (0)	0.95	0.04	0.01
Retired (1)	0.00	0.90	0.10
Dead (2)	0.00	0.00	1.00

Under the guidelines of TU's employee benefits plan, a pension benefit of 25,000 will be paid on each birthday while she is retired. This pension benefit expires the day before Jennifer's 54th birthday.

Using $v = 0.90$, calculate the actuarial present value of Jennifer's employee benefits plan.

8. You are given a three-state model and the following transition intensities for (x) for all t :

- (i) $\mu_x^{01}(t) = 0.022$; $\mu_x^{02}(t) = 0.027$
(ii) $\mu_x^{12}(t) = 0.012$; $\mu_x^{10}(t) = 0.010$
(iii) All other transition intensities are equal to 0.

Write the Kolmogorov forward differential equations with the associated boundary conditions for this model for: (i) ${}_t p_x^{01}$ and (ii) ${}_t p_x^{11}$.

9. For a multi-state model with 3 possible states, an employee of the TU Company can be in one of the following states at the beginning of each year: Active (state 0), Disabled (state 1), Dead (state 2).

A currently Active TU employee, Anne, just turned 60 years old. You are given the following table of her annual transition probabilities for $k = 0, 1, 2, 3$.

State i	p_{60+k}^{i0}	p_{60+k}^{i1}	p_{60+k}^{i2}
Active (0)	0.95	0.04	0.01
Disabled (1)	0.00	0.90	0.10
Dead (2)	0.00	0.00	1.00

She has just purchased a 3-year fully discrete term insurance with a death benefit of 25,000. She will pay a premium at the beginning of each year as long as she is Active. Using $v = 0.95$, calculate the level annual net premium for this policy.

10. Jerry (55)'s future lifetime follows the Standard Sickness-Death Model. He is currently Sick. He just purchased a special whole life policy that has the following features:
- (i) The policy pays a death benefit of 20,000 at the moment of death.
 - (ii) The policy pays a continuous sickness benefit at a rate of 2,500 per year while the insured is in the Sick state.
 - (iii) The policy will provide a 20-year certain annuity-immediate at the end of 10 years if the insured is Healthy at time 10. This annuity will make annual payments of 1,200.

Using $i = 0.05$, calculate the actuarial present value of this policy.

11. You are given the following matrix containing annual Markov transition probabilities with states 0, 1, 2 for (x). State (0) represents Healthy, state (1) represents Disabled, and state (2) represents Dead.

$$\begin{bmatrix} 0.82 & 0.15 & 0.03 \\ 0.14 & 0.81 & 0.05 \\ 0 & 0 & 1 \end{bmatrix}$$

- (a) Calculate ${}_3p_x^{12}$.
- (b) There are seven Healthy lives age (x) with independent future lifetimes. Calculate the probability that exactly three of them will be Disabled at the end of 2 years.

12. You are given a three-state model where:

State 0 represents Healthy
 State 1 represents Disabled
 State 2 represents Dead

You are given the following transition intensities for all t :

- (i) $\mu_{x+t}^{01} = 0.025$

(ii) $\mu_{x+t}^{02} = 0.030$

(iii) $\mu_{x+t}^{12} = 0.045$

All other transition intensities are equal to 0.

Calculate ${}_5p_x^{02} + {}_5p_x^{12}$.

13. You are given the following matrix containing annual Markov transition probabilities with states 0, 1, 2 for a machine (x).

State (0) represents Operational, state (1) represents Under Repair, and state (2) represents Unrepairable.

$$\begin{bmatrix} 0.84 & 0.15 & 0.01 \\ 0.17 & 0.78 & 0.05 \\ 0 & 0 & 1 \end{bmatrix}$$

There are 10 operational machines age (x) with independent future lifetimes. Calculate the probability that 7 will be unrepairable by the end of 3 years.

14. You are given a three-state model about (x) where State 0 represents Healthy, State 1 represents Disabled, and State 2 represents Dead.

You are given the following transition intensities for all t:

(i) $\mu_x^{01}(t) = 0.046$; $\mu_x^{02}(t) = 0.038$; $\mu_x^{12}(t) = 0.062$

(ii) All other transition intensities are equal to 0.

(x) is currently Healthy. She purchases a whole life disability policy which will pay a continuous rate of 6,000 per year while she is disabled. You are given $\delta = 0.03$. She will pay for this policy with continuous net annual premiums of P, payable while she is healthy.

Calculate P.

Answers

1. 68,750

2. 0.9546005

3. 0.007934667

4. (a)

$$(i) \frac{d}{dt} {}_t p_x^{00} = 0.016 {}_t p_x^{01} - 0.033 {}_t p_x^{00}$$

with boundary conditions ${}_0 p_x^{00} = 1, {}_0 p_x^{01} = 0$

$$(ii) \frac{d}{dt} {}_t p_x^{01} = 0.015 {}_t p_x^{00} - 0.04 {}_t p_x^{01}$$

with boundary conditions ${}_0 p_x^{00} = 1, {}_0 p_x^{01} = 0$

(b) 0.983583

5. 39,186.99

$$6. \begin{bmatrix} 0.6538 & 0.1821 & 0.1641 \\ 0 & 0.8086 & 0.1914 \\ 0 & 0 & 1 \end{bmatrix}$$

7. 4,270.21

$$8. (i) \frac{d}{dt} {}_t p_x^{01} = {}_t p_x^{00}(0.022) - {}_t p_x^{01}(0.01 + 0.012)$$

with boundary conditions ${}_0 p_x^{01} = 0$ and ${}_0 p_x^{00} = 1$

$$(ii) \frac{d}{dt} {}_t p_x^{11} = {}_t p_x^{10}(0.022) - {}_t p_x^{11}(0.01 + 0.012)$$

with boundary conditions ${}_0 p_x^{11} = 1$ and ${}_0 p_x^{10} = 0$

9. 329.09

10. 37,443.38

11. (a) 0.1354 (b) 0.167

12. $0.14303 + 0.20148 = 0.3445$ 13. $4.1 * 10^{-8}$

14. 3,000